KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3261 | 3 Credits: 4 | Program Core | Source-grounded

# Fundamentals of Network

 To provide basic concepts of Computer Networks from a Network Administrator’s

OFFICIAL SOURCE DECLARATION

## How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3261 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 3261.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

## Official course information

**Course information**

| Item | Official information |
|---|---|
| Programme | Diploma in Information Technology |
| Course code | 3261 |
| Course title | Fundamentals of Network |
| Semester | 3 Credits: 4 |
| Course category | Program Core |
| Periods per week | 4 (L:4 T:0 P:0) Periods per semester: 60 |

| Item | Official information |
| --- | --- |
| Periods per semester | 60 |
| Credits | 4 |
| Prerequisite | See the official syllabus PDF |
| Assessment | Use the official assessment section reproduced below |

### Modules

4 official module sections detected.

### Topic entries

39 source-aligned topic entries retained.

### Study mode

Theory and application emphasis.

STUDY METHOD

## How to use this lesson

### First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

### Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

### Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

### Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

# Objectives, outcomes and mapping

## Course objectives

- To provide basic concepts of Computer Networks from a Network Administrator’s
- point-of-view.

#### Course outcomes

| CO | Official outcome | Study level |
|---|---|---|
| CO1 | 15 Understanding its Physical Layer. Illustrate basic concepts of Local Area Networks, IP | See official syllabus |
| CO2 | 14 Understanding Addressing and Subnetting. Summarize basic concepts of Routing Algorithms and | See official syllabus |
| CO3 | 15 Understanding Routing Protocols. Illustrate basic concepts of Transport layer protocols, | See official syllabus |
| CO4 | 14 Understanding Application layer protocols and Network Security. | See official syllabus |

#### CO–PO mapping evidence

| Row | Extracted official mapping |
|---|---|
| CO1 | CO1 2 |
| CO2 | CO2 2 |
| CO3 | CO3 2 |
| CO4 | CO4 2 |

COVERAGE AUDIT

## Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

**Module coverage**

| Module | Official heading | Topic entries | Hours |
|---|---|---|---|
| Module 1 | Outline the basic concepts of Computer Networks and its Physical Layer. | 7 | As specified |
| Module 2 | Subnetting. | 13 | As specified |
| Module 3 | Summarize basic concept of Routing Algorithms and Routing Protocols | 8 | As specified |
| Module 4 | protocols and Network Security. | 11 | As specified |

MODULE 1

# Outline the basic concepts of Computer Networks and its Physical Layer.

This module is presented from the official syllabus scope for Fundamentals of Network. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

### Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: Outline the basic concepts of Computer Networks and its Physical Layer.
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

## elements of a computer network and 2 Understanding Network Topologies

elements of a computer network and 2 Understanding Network Topologies is an official topic in Module 1 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify elements of a computer network and 2 Understanding Network Topologies using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, elements of a computer network and 2 understanding network topologies should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What elements of a computer network and 2 Understanding Network Topologies means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

> **Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 1-□□ elements of a computer network and 2 Understanding Network Topologies □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

## Illustrate Transmission Mediums.

Illustrate Transmission Mediums. is an official topic in Module 1 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Illustrate Transmission Mediums. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, illustrate transmission mediums. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Illustrate Transmission Mediums. means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ Illustrate Transmission Mediums. □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

## Outline Types of Networks

Outline Types of Networks is an official topic in Module 1 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Outline Types of Networks using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, outline types of networks should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Outline Types of Networks means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

> **Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ Outline Types of Networks □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

## Summarize Switched Networks 2 Understanding Illustrate the duties of each layer in OSI

Summarize Switched Networks 2 Understanding Illustrate the duties of each layer in OSI is an official topic in Module 1 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarize Switched Networks 2 Understanding Illustrate the duties of each layer in OSI using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, summarize switched networks 2 understanding illustrate the duties of each layer in osi should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Summarize Switched Networks 2 Understanding Illustrate the duties of each layer in OSI means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ Summarize Switched Networks 2 Understanding Illustrate the duties of each layer in OSI □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

## 3 Understanding Model Outline the duties of each layer in

3 Understanding Model Outline the duties of each layer in is an official topic in Module 1 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 3 Understanding Model Outline the duties of each layer in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 3 understanding model outline the duties of each layer in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 3 Understanding Model Outline the duties of each layer in means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 1-[illegible] 3 Understanding Model Outline the duties of each layer in [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## 2 Understanding TCP/IP Model

2 Understanding TCP/IP Model is an official topic in Module 1 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding TCP/IP Model using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding tcp/ip model should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 2 Understanding TCP/IP Model means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ 2 Understanding TCP/IP Model □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

**Summarize Network Devices. 2 Understanding Computer Network – Basic Elements – Network Topologies (Bus, Star, Ring and Mesh) – Transmission Medium (Twisted Pair, Co-axial, Fiber Optic and Electromagnetic Waves) – Types of Networks (LAN, MAN, WAN, PAN and Wireless) – Switched Networks (Circuit Switched, Packet Switched) Protocol Stack – OSI Model – Functions of each layer – TCP/IP Model – Functions of Each Layer – Difference between OSI and TCP/IP models Networking Devices (NIC, Hub, Modem, Bridge, Switch, Router, Gateway and Firewall) Illustrate basic concepts of Local Area Networks, IP Addressing and**

Summarize Network Devices. 2 Understanding Computer Network – Basic Elements – Network Topologies (Bus, Star, Ring and Mesh) – Transmission Medium (Twisted Pair, Co-axial, Fiber Optic and Electromagnetic Waves) – Types of Networks (LAN, MAN, WAN, PAN and Wireless) – Switched Networks (Circuit Switched, Packet Switched) Protocol Stack – OSI Model – Functions of each layer – TCP/IP Model – Functions of Each Layer – Difference between OSI and TCP/IP models Networking Devices (NIC, Hub, Modem, Bridge, Switch, Router, Gateway and Firewall) Illustrate basic concepts of Local Area Networks, IP Addressing and is an official topic in Module 1 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarize Network Devices. 2 Understanding Computer Network – Basic Elements – Network Topologies (Bus, Star, Ring and Mesh) – Transmission Medium (Twisted Pair, Co-axial, Fiber Optic and Electromagnetic Waves) – Types of Networks (LAN, MAN, WAN, PAN and Wireless) – Switched Networks (Circuit Switched, Packet Switched) Protocol Stack – OSI Model – Functions of each layer – TCP/IP Model – Functions of Each Layer – Difference between OSI and TCP/IP models Networking Devices (NIC, Hub, Modem, Bridge, Switch, Router, Gateway and Firewall) Illustrate basic concepts of Local Area Networks, IP Addressing and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, summarize network devices. 2 understanding computer network – basic elements – network topologies (bus, star, ring and mesh) – transmission medium (twisted pair, co-axial, fiber optic and electromagnetic waves) – types of networks (lan, man, wan, pan and wireless) – switched networks (circuit switched, packet switched) protocol stack – osi model – functions of each layer – tcp/ip model – functions of each layer – difference between osi and tcp/ip models networking devices (nic, hub, modem, bridge, switch, router, gateway and firewall) illustrate basic concepts of local area networks, ip addressing and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What Summarize Network Devices. 2 Understanding Computer Network – Basic Elements – Network Topologies (Bus, Star, Ring and Mesh) – Transmission Medium (Twisted Pair, Co-axial, Fiber Optic and Electromagnetic Waves) – Types of Networks (LAN, MAN, WAN, PAN and Wireless) – Switched Networks (Circuit Switched, Packet Switched) Protocol Stack – OSI Model – Functions of each layer – TCP/IP Model – Functions of Each Layer – Difference between OSI and TCP/IP models Networking Devices (NIC, Hub, Modem, Bridge, Switch, Router, Gateway and Firewall) Illustrate basic concepts of Local Area Networks, IP Addressing and means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ Summarize Network Devices. 2 Understanding Computer Network – Basic Elements – Network Topologies (Bus, Star, Ring and Mesh) – Transmission Medium (Twisted Pair, Co-axial, Fiber Optic and Electromagnetic Waves) – Types of Networks (LAN, MAN, WAN, PAN and Wireless) – Switched Networks (Circuit Switched, Packet Switched) Protocol Stack – OSI Model – Functions of each layer – TCP/IP Model – Functions of Each Layer – Difference between OSI and TCP/IP models Networking Devices (NIC, Hub, Modem, Bridge, Switch, Router, Gateway and Firewall) Illustrate basic concepts of Local Area Networks, IP Addressing and □□□□□□□□□□□ □□□□□ definition/meaning

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### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Outline the basic concepts of Computer Networks and its Physical Layer.
                 Outline Computer Network, Basic
   M1.01         elements of a computer network and             2
Understanding
                 Network Topologies
   M1.02         Illustrate Transmission Mediums.               2
Understanding
   M1.03         Outline Types of Networks                      2
Understanding
   M1.04         Summarize Switched Networks                    2
Understanding
                 Illustrate the duties of each layer in OSI
   M1.05                                                        3
Understanding
                 Model
                 Outline the duties of each layer in
   M1.06                                                        2
Understanding
                 TCP/IP Model
   M1.07         Summarize Network Devices.                     2
Understanding
Contents:
Computer Network – Basic Elements – Network Topologies (Bus, Star, Ring and Mesh)
–
Transmission Medium (Twisted Pair, Co-axial, Fiber Optic and Electromagnetic
Waves) –
Types of Networks (LAN, MAN, WAN, PAN and Wireless) – Switched Networks (Circuit
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

# Subnetting.

This module is presented from the official syllabus scope for Fundamentals of Network. Use the topic cards for learning and the source transcript for exact coverage.

| **As specified** | **CO-linked** |
| --- | --- |
| official hours | see outcomes |

## Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Subnetting.
- Official duration: As specified
- Official topic entries captured: 13
- The full source excerpt is preserved below for coverage checking.

## 1 Understanding Techniques

1 Understanding Techniques is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 1 Understanding Techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 1 understanding techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 1 Understanding Techniques means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-[illegible] 1 Understanding Techniques [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## Summarize Ethernet and its types

Summarize Ethernet and its types is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarize Ethernet and its types using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, summarize ethernet and its types should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Summarize Ethernet and its types means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-[illegible] Summarize Ethernet and its types [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## Outline ARP and RARP

Outline ARP and RARP is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Outline ARP and RARP using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, outline arp and rarp should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What Outline ARP and RARP means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-[illegible] Outline ARP and RARP [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## Summarize IP Address and IP 1 Understanding Addressing Outline Classful IP Addressing and IP

Summarize IP Address and IP 1 Understanding Addressing Outline Classful IP Addressing and IP is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarize IP Address and IP 1 Understanding Addressing Outline Classful IP Addressing and IP using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, summarize ip address and ip 1 understanding addressing outline classful ip addressing and ip should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Summarize IP Address and IP 1 Understanding Addressing Outline Classful IP Addressing and IP means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ Summarize IP Address and IP 1 Understanding Addressing Outline Classful IP Addressing and IP □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

## 1 Understanding Address Classes.

1 Understanding Address Classes. is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 1 Understanding Address Classes. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 1 understanding address classes. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 1 Understanding Address Classes. means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ 1 Understanding Address Classes. □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

## Summarize IPv4 Addressing Types

Summarize IPv4 Addressing Types is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarize IPv4 Addressing Types using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, summarize ipv4 addressing types should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Summarize IPv4 Addressing Types means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-[illegible] Summarize IPv4 Addressing Types [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## Summarize IPv4 Reserved Addresses 1 Understanding Illustrate Packet Flow in an IPv4

Summarize IPv4 Reserved Addresses 1 Understanding Illustrate Packet Flow in an IPv4 is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarize IPv4 Reserved Addresses 1 Understanding Illustrate Packet Flow in an IPv4 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, summarize ipv4 reserved addresses 1 understanding illustrate packet flow in an ipv4 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Summarize IPv4 Reserved Addresses 1 Understanding Illustrate Packet Flow in an IPv4 means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ Summarize IPv4 Reserved Addresses 1 Understanding Illustrate Packet Flow in an IPv4 □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

## 1 Understanding Network

1 Understanding Network is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 1 Understanding Network using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 1 understanding network should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 1 Understanding Network means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ 1 Understanding Network □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

## Summarize IPv4 Fragmentation 1 Understanding Outline Limitations of Classful

Summarize IPv4 Fragmentation 1 Understanding Outline Limitations of Classful is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarize IPv4 Fragmentation 1 Understanding Outline Limitations of Classful using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, summarize ipv4 fragmentation 1 understanding outline limitations of classful should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What Summarize IPv4 Fragmentation 1 Understanding Outline Limitations of Classful means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ Summarize IPv4 Fragmentation 1 Understanding Outline Limitations of Classful □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

## 1 Understanding Addressing

1 Understanding Addressing is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 1 Understanding Addressing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 1 understanding addressing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 1 Understanding Addressing means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ 1 Understanding Addressing □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

## Illustrate Subnet Mask and Subnetting 2 Understanding Summarize Supernetting and Classless

Illustrate Subnet Mask and Subnetting 2 Understanding Summarize Supernetting and Classless is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Illustrate Subnet Mask and Subnetting 2 Understanding Summarize Supernetting and Classless using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, illustrate subnet mask and subnetting 2 understanding summarize supernetting and classless should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Illustrate Subnet Mask and Subnetting 2 Understanding Summarize Supernetting and Classless means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ Illustrate Subnet Mask and Subnetting 2 Understanding Summarize Supernetting and Classless □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

## 1 Understanding Inter-Domain Routing

1 Understanding Inter-Domain Routing is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 1 Understanding Inter-Domain Routing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 1 understanding inter-domain routing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 1 Understanding Inter-Domain Routing means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ 1 Understanding Inter-Domain Routing □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

**Outline Variable Length Subnet Mask 1 Understanding Local Area Network – Types of Network Links – Medium Access Control Techniques – Random Access Techniques (ALOHA, CSMA, CSMA/CD, CSMA/CA) – Ethernet – Types of Ethernet ARP and RARP – Address Resolution Protocol – Reverse Address Resolution Protocol IP Addressing – IP Address – IPv4 – IPv6 – Classful Addressing – IPv4 Address Classes (Class A, B, C, D and E) – IPv4 Addressing Types (Unicast, Broadcast and Multicast) – IPv4 Reserved Address (Loopback, Zero and Private Addresses) – Packet Flow in IPv4 Network – IPv4 Fragmentation – Limitations of Classful Addressing – Subnet Mask – Subnetting – Supernetting – Classless Inter-Domain Routing – Variable Length Subnet Mask**

Outline Variable Length Subnet Mask 1 Understanding Local Area Network – Types of Network Links – Medium Access Control Techniques – Random Access Techniques (ALOHA, CSMA, CSMA/CD, CSMA/CA) – Ethernet – Types of Ethernet ARP and RARP – Address Resolution Protocol – Reverse Address Resolution Protocol IP Addressing – IP Address – IPv4 – IPv6 – Classful Addressing – IPv4 Address Classes (Class A, B, C, D and E) – IPv4 Addressing Types (Unicast, Broadcast and Multicast) – IPv4 Reserved Address (Loopback, Zero and Private Addresses) – Packet Flow in IPv4 Network – IPv4 Fragmentation – Limitations of Classful Addressing – Subnet Mask – Subnetting – Supernetting – Classless Inter-Domain Routing – Variable Length Subnet Mask is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Outline Variable Length Subnet Mask 1 Understanding Local Area Network – Types of Network Links – Medium Access Control Techniques – Random Access Techniques (ALOHA, CSMA, CSMA/CD, CSMA/CA) – Ethernet – Types of Ethernet ARP and RARP – Address Resolution Protocol – Reverse Address Resolution Protocol IP Addressing – IP Address – IPv4 – IPv6 – Classful Addressing – IPv4 Address Classes (Class A, B, C, D and E) – IPv4 Addressing Types (Unicast, Broadcast and Multicast) – IPv4 Reserved Address (Loopback, Zero and Private Addresses) – Packet Flow in IPv4 Network – IPv4 Fragmentation – Limitations of Classful Addressing – Subnet Mask – Subnetting – Supernetting – Classless Inter-

Domain Routing – Variable Length Subnet Mask using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, outline variable length subnet mask 1 understanding local area network – types of network links – medium access control techniques – random access techniques (aloha, csma, csma/cd, csma/ca) – ethernet – types of ethernet arp and rarp – address resolution protocol – reverse address resolution protocol ip addressing – ip address – ipv4 – ipv6 – classful addressing – ipv4 address classes (class a, b, c, d and e) – ipv4 addressing types (unicast, broadcast and multicast) – ipv4 reserved address (loopback, zero and private addresses) – packet flow in ipv4 network – ipv4 fragmentation – limitations of classful addressing – subnet mask – subnetting – supernetting – classless inter-domain routing – variable length subnet mask should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Outline Variable Length Subnet Mask 1 Understanding Local Area Network – Types of Network Links – Medium Access Control Techniques – Random Access Techniques (ALOHA, CSMA, CSMA/CD, CSMA/CA) – Ethernet – Types of Ethernet ARP and RARP – Address Resolution Protocol – Reverse Address Resolution Protocol IP Addressing – IP Address – IPv4 – IPv6 – Classful Addressing – IPv4 Address Classes (Class A, B, C, D and E) – IPv4 Addressing Types (Unicast, Broadcast and Multicast) – IPv4 Reserved Address (Loopback, Zero and Private Addresses) – Packet Flow in IPv4 Network – IPv4 Fragmentation – Limitations of Classful Addressing – Subnet Mask – Subnetting – Supernetting – Classless Inter-Domain Routing – Variable Length Subnet Mask means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ Outline Variable Length Subnet Mask 1 Understanding Local Area Network – Types of Network Links – Medium Access

Control Techniques – Random Access Techniques (ALOHA, CSMA, CSMA/CD, CSMA/CA) – Ethernet – Types of Ethernet ARP and RARP – Address Resolution Protocol – Reverse Address Resolution Protocol IP Addressing – IP Address – IPv4 – IPv6 – Classful Addressing – IPv4 Address Classes (Class A, B, C, D and E) – IPv4 Addressing Types (Unicast, Broadcast and Multicast) – IPv4 Reserved Address (Loopback, Zero and Private Addresses) – Packet Flow in IPv4 Network – IPv4 Fragmentation – Limitations of Classful Addressing – Subnet Mask – Subnetting – Supernetting – Classless Inter-Domain Routing – Variable Length Subnet Mask □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□.

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Subnetting.
              Summarize Medium Access Control
   M2.01                                                        1
Understanding
              Techniques
   M2.02      Summarize Ethernet and its types                  1
Understanding
   M2.03      Outline ARP and RARP                              1
Understanding

   M2.04      Summarize         IP   Address     and     IP      1
Understanding
              Addressing

              Outline Classful IP Addressing and IP
   M2.05                                                  1
Understanding
              Address Classes.
   M2.06      Summarize IPv4 Addressing Types             1
Understanding
   M2.07      Summarize IPv4 Reserved Addresses           1
Understanding
              Illustrate Packet Flow in an IPv4
   M2.08                                                  1
Understanding
              Network
   M2.09      Summarize IPv4 Fragmentation                1
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

# Summarize basic concept of Routing Algorithms and Routing Protocols

This module is presented from the official syllabus scope for Fundamentals of Network. Use the topic cards for learning and the source transcript for exact coverage.

| **As specified** | **CO-linked** |
|---|---|
| official hours | see outcomes |

### Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: Summarize basic concept of Routing Algorithms and Routing Protocols
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

## 2 Understanding Algorithms Summarize Classification of Routing

2 Understanding Algorithms Summarize Classification of Routing is an official topic in Module 3 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding Algorithms Summarize Classification of Routing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding algorithms summarize classification of routing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 2 Understanding Algorithms Summarize Classification of Routing means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 3-[illegible] 2 Understanding Algorithms Summarize Classification of Routing [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## 2 Understanding Algorithms

2 Understanding Algorithms is an official topic in Module 3 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding Algorithms using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding algorithms should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 2 Understanding Algorithms means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-[illegible] 2 Understanding Algorithms [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## Outline Routing Algorithm Metrics

Outline Routing Algorithm Metrics is an official topic in Module 3 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Outline Routing Algorithm Metrics using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, outline routing algorithm metrics should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Outline Routing Algorithm Metrics means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-[illegible] Outline Routing Algorithm Metrics [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## Summarize Internet Architecture 1 Understanding Outline Routing Protocols and its

Summarize Internet Architecture 1 Understanding Outline Routing Protocols and its is an official topic in Module 3 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarize Internet Architecture 1 Understanding Outline Routing Protocols and its using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, summarize internet architecture 1 understanding outline routing protocols and its should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Summarize Internet Architecture 1 Understanding Outline Routing Protocols and its means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-[illegible] Summarize Internet Architecture 1 Understanding Outline Routing Protocols and its [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## 2 Understanding classifications. Summarize Routing Information

2 Understanding classifications. Summarize Routing Information is an official topic in Module 3 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding classifications. Summarize Routing Information using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding classifications. summarize routing information should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 2 Understanding classifications. Summarize Routing Information means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 3-□□ 2 Understanding classifications. Summarize Routing Information □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

## 2 Understanding Protocol Illustrate Open Shortest Path First

2 Understanding Protocol Illustrate Open Shortest Path First is an official topic in Module 3 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding Protocol Illustrate Open Shortest Path First using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding protocol illustrate open shortest path first should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 2 Understanding Protocol Illustrate Open Shortest Path First means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 3-[illegible] 2 Understanding Protocol Illustrate Open Shortest Path First [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## 2 Understanding Routing Protocol

2 Understanding Routing Protocol is an official topic in Module 3 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding Routing Protocol using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding routing protocol should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 2 Understanding Routing Protocol means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-[illegible] 2 Understanding Routing Protocol [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

**Summarize Border Gateway Protocol 2 Understanding IP Routing – Routing Algorithm – Classification (Static vs Dynamic, Single-Path vs Multi- Path, Intra-Domain vs Inter-Domain, Link-State vs Distance Vector, Host Intelligent vs Router Intelligent) – Routing Algorithm Metrics (Path Length, Delay, Bandwidth, Load, Communication Cost and Reliability) – Internet Architecture Routing Protocols – Classification – Routing Information Protocol – Open Shortest Path First – Border Gateway Protocol. Illustrate basic concepts of Transport layer protocols, Application layer**

Summarize Border Gateway Protocol 2 Understanding IP Routing – Routing Algorithm – Classification (Static vs Dynamic, Single-Path vs Multi- Path, Intra-Domain vs Inter-Domain, Link-State vs Distance Vector, Host Intelligent vs Router Intelligent) – Routing Algorithm Metrics (Path Length, Delay, Bandwidth, Load, Communication Cost and Reliability) – Internet Architecture Routing Protocols – Classification – Routing Information Protocol – Open Shortest Path First – Border Gateway Protocol. Illustrate basic concepts of Transport layer protocols, Application layer is an official topic in Module 3 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarize Border Gateway Protocol 2 Understanding IP Routing – Routing Algorithm – Classification (Static vs Dynamic, Single-Path vs Multi- Path, Intra-Domain vs Inter-Domain, Link-State vs Distance Vector, Host Intelligent vs Router Intelligent) – Routing Algorithm Metrics (Path Length, Delay, Bandwidth, Load, Communication Cost and Reliability) – Internet Architecture Routing Protocols – Classification – Routing Information Protocol – Open Shortest Path First – Border Gateway Protocol. Illustrate basic concepts of Transport layer protocols, Application layer using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, summarize border gateway protocol 2 understanding ip routing – routing algorithm – classification (static vs dynamic, single-path vs multi- path, intra-domain vs inter-domain, link-state vs distance vector, host intelligent vs router intelligent) – routing algorithm metrics (path length, delay, bandwidth, load, communication cost and reliability) – internet architecture routing protocols – classification – routing information protocol – open shortest path first – border gateway protocol. illustrate basic concepts of transport layer protocols, application layer should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Summarize Border Gateway Protocol 2 Understanding IP Routing – Routing Algorithm – Classification (Static vs Dynamic, Single-Path vs Multi- Path, Intra-Domain vs Inter-Domain, Link-State vs Distance Vector, Host Intelligent vs Router Intelligent) – Routing Algorithm Metrics (Path Length, Delay, Bandwidth, Load, Communication Cost and Reliability) – Internet Architecture Routing Protocols – Classification – Routing Information Protocol – Open Shortest Path First – Border Gateway Protocol. Illustrate basic concepts of Transport layer protocols, Application layer means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ Summarize Border Gateway Protocol 2 Understanding IP Routing – Routing Algorithm – Classification (Static vs Dynamic, Single-Path vs Multi- Path, Intra-Domain vs Inter-Domain, Link-State vs Distance Vector, Host Intelligent vs Router Intelligent) – Routing Algorithm Metrics (Path Length, Delay, Bandwidth, Load, Communication Cost and Reliability) – Internet Architecture Routing Protocols – Classification – Routing Information Protocol – Open Shortest Path First – Border Gateway Protocol. Illustrate basic concepts of Transport layer protocols, Application layer □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Summarize basic concept of Routing Algorithms and Routing Protocols

              Illustrate Network Routing and Routing
   M3.01                                                  2
Understanding
              Algorithms
              Summarize Classification of Routing
   M3.02                                                  2
Understanding
              Algorithms
   M3.03      Outline Routing Algorithm Metrics           2
Understanding
   M3.04      Summarize Internet Architecture             1
Understanding
              Outline Routing Protocols and its
   M3.05                                                  2
Understanding
              classifications.
              Summarize Routing Information
   M3.06                                                  2
Understanding
              Protocol
              Illustrate Open Shortest Path First
   M3.07                                                  2
Understanding
              Routing Protocol
   M3.08       Summarize Border Gateway Protocol              2
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

## protocols and Network Security.

This module is presented from the official syllabus scope for Fundamentals of Network. Use the topic cards for learning and the source transcript for exact coverage.

| **As specified** | **CO-linked** |
| --- | --- |
| official hours | see outcomes |

## Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: protocols and Network Security.
- Official duration: As specified
- Official topic entries captured: 11
- The full source excerpt is preserved below for coverage checking.

## Illustrate Ports and Sockets 1 Understanding Outline Transmission Control Protocol

Illustrate Ports and Sockets 1 Understanding Outline Transmission Control Protocol is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Illustrate Ports and Sockets 1 Understanding Outline Transmission Control Protocol using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, illustrate ports and sockets 1 understanding outline transmission control protocol should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Illustrate Ports and Sockets 1 Understanding Outline Transmission Control Protocol means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ Illustrate Ports and Sockets 1 Understanding Outline Transmission Control Protocol □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

## 2 Understanding and User Datagram Protocols

2 Understanding and User Datagram Protocols is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding and User Datagram Protocols using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding and user datagram protocols should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 2 Understanding and User Datagram Protocols means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ 2 Understanding and User Datagram Protocols □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

## Summarize Hypertext Transfer Protocol 2 Understanding Outline Simple Mail Transfer Control

Summarize Hypertext Transfer Protocol 2 Understanding Outline Simple Mail Transfer Control is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarize Hypertext Transfer Protocol 2 Understanding Outline Simple Mail Transfer Control using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, summarize hypertext transfer protocol 2 understanding outline simple mail transfer control should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Summarize Hypertext Transfer Protocol 2 Understanding Outline Simple Mail Transfer Control means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ Summarize Hypertext Transfer Protocol 2 Understanding Outline Simple Mail Transfer Control □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

## 1 Understanding Protocol

1 Understanding Protocol is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 1 Understanding Protocol using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 1 understanding protocol should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 1 Understanding Protocol means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-[illegible] 1 Understanding Protocol [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## Illustrate File Transfer Protocol

Illustrate File Transfer Protocol is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Illustrate File Transfer Protocol using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, illustrate file transfer protocol should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What Illustrate File Transfer Protocol means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ Illustrate File Transfer Protocol □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

## Summarize Domain Name Server 1 Understanding Outline Dynamic Host Configuration

Summarize Domain Name Server 1 Understanding Outline Dynamic Host Configuration is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarize Domain Name Server 1 Understanding Outline Dynamic Host Configuration using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, summarize domain name server 1 understanding outline dynamic host configuration should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Summarize Domain Name Server 1 Understanding Outline Dynamic Host Configuration means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ Summarize Domain Name Server 1 Understanding Outline Dynamic Host Configuration □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

## 1 Understanding Protocol Summarize Network Security Needs,

1 Understanding Protocol Summarize Network Security Needs, is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 1 Understanding Protocol Summarize Network Security Needs, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 1 understanding protocol summarize network security needs, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 1 Understanding Protocol Summarize Network Security Needs, means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 4-□□ 1 Understanding Protocol Summarize Network Security Needs, □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

## 2 Understanding History and Four Pillars Illustrate Network Security Terms and

2 Understanding History and Four Pillars Illustrate Network Security Terms and is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding History and Four Pillars Illustrate Network Security Terms and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding history and four pillars illustrate network security terms and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 2 Understanding History and Four Pillars Illustrate Network Security Terms and means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-[illegible] 2 Understanding History and Four Pillars Illustrate Network Security Terms and [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## 1 Understanding Security Components Summarize Types of Attacks and

1 Understanding Security Components Summarize Types of Attacks and is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 1 Understanding Security Components Summarize Types of Attacks and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 1 understanding security components summarize types of attacks and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 1 Understanding Security Components Summarize Types of Attacks and means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

> **Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 4-□□ 1 Understanding Security Components Summarize Types of Attacks and □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

## 1 Understanding Known Security Attacks

1 Understanding Known Security Attacks is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 1 Understanding Known Security Attacks using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 1 understanding known security attacks should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 1 Understanding Known Security Attacks means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-[illegible] 1 Understanding Known Security Attacks [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

**Outline Firewalls 1 Understanding TCP and UDP – Ports – Sockets – User Datagram Protocols – Transmission Control Protocol – Compare TCP and UDP Application Layer Protocols – Hypertext Transfer Protocol – Simple Mail Transfer Control Protocol – File Transfer Protocol – Domain Name Server – Dynamic Host Configuration Protocol Network Security – Needs – History – Four Pillars – Security Terms – Security Components – Types of Attacks – Known Security Attacks – Firewalls Text / Reference: T/R Book Title / Author Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. Elements of T1 Computer Networking: An Integrated Approach. CareerMonk Publications, 2014. Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition, R1 Sybex (Wiley) Publication, 2016 Online Resources: Sl.No. Website Link https://www.tutorialspoint.com/data_communication_computer_network/index.h 1 tm 2 https://www.javatpoint.com/computer-network-tutorial 3 https://www.geeksforgeeks.org/computer-network-tutorials/**

Outline Firewalls 1 Understanding TCP and UDP – Ports – Sockets – User Datagram Protocols – Transmission Control Protocol – Compare TCP and UDP Application Layer Protocols – Hypertext Transfer Protocol – Simple Mail Transfer Control Protocol – File Transfer Protocol – Domain Name Server – Dynamic Host Configuration Protocol Network Security – Needs – History – Four Pillars – Security Terms – Security Components – Types of Attacks – Known Security Attacks – Firewalls Text / Reference: T/R Book Title / Author Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. Elements of T1 Computer Networking: An Integrated Approach. CareerMonk Publications, 2014. Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition, R1 Sybex (Wiley) Publication, 2016 Online Resources: Sl.No. Website Link https://www.tutorialspoint.com/data_communication_computer_network/index.h 1 tm 2 https://www.javatpoint.com/computer-network-tutorial 3 https://www.geeksforgeeks.org/computer-network-tutorials/ is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Outline Firewalls 1 Understanding TCP and UDP – Ports – Sockets – User Datagram Protocols – Transmission Control Protocol – Compare TCP and UDP Application Layer Protocols – Hypertext Transfer Protocol – Simple Mail Transfer Control Protocol – File Transfer Protocol – Domain Name Server – Dynamic Host Configuration Protocol Network Security – Needs – History – Four Pillars – Security Terms – Security Components – Types of Attacks – Known Security Attacks – Firewalls Text / Reference: T/R Book Title / Author Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. Elements of T1 Computer Networking: An Integrated Approach. CareerMonk Publications, 2014. Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition, R1 Sybex (Wiley) Publication, 2016 Online Resources: Sl.No. Website Link https://www.tutorialspoint.com/data_communication_computer_network/index.h 1 tm 2 https://www.javatpoint.com/computer-network-tutorial 3 https://www.geeksforgeeks.org/computer-network-tutorials/ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, outline firewalls 1 understanding tcp and udp – ports – sockets – user datagram protocols – transmission control protocol – compare tcp and udp application layer protocols – hypertext transfer protocol – simple mail transfer control protocol – file transfer protocol – domain name server – dynamic host configuration protocol network security – needs – history – four pillars – security terms – security components – types of attacks – known security attacks – firewalls text / reference: t/r book title / author narasimha karumanchi, dr. a. damodaram and dr. m. sreenivasa rao. elements of t1 computer networking: an integrated approach. careermonk publications, 2014. todd lammle, ccna routing and switching complete study guide, 2nd edition, r1 sybex (wiley) publication, 2016 online resources: sl.no. website link https://www.tutorialspoint.com/data_communication_computer_network/index.h 1 tm 2 https://www.javatpoint.com/computer-network-tutorial 3 https://www.geeksforgeeks.org/computer-network-tutorials/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Outline Firewalls 1 Understanding TCP and UDP – Ports – Sockets – User Datagram Protocols – Transmission Control Protocol – Compare TCP and UDP Application Layer Protocols – Hypertext Transfer Protocol – Simple Mail Transfer Control Protocol – File Transfer Protocol – Domain Name Server – Dynamic Host Configuration Protocol Network Security – Needs – History – Four Pillars – |

| Aspect | What to prepare |
| --- | --- |
| | Security Terms – Security Components – Types of Attacks – Known Security Attacks – Firewalls Text / Reference: T/R Book Title / Author Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. Elements of T1 Computer Networking: An Integrated Approach. CareerMonk Publications, 2014. Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition, R1 Sybex (Wiley) Publication, 2016 Online Resources: Sl.No. Website Link https://www.tutorialspoint.com/data_communication_computer_network/index.h 1 tm 2 https://www.javatpoint.com/computer-network-tutorial 3 https://www.geeksforgeeks.org/computer-network-tutorials/ means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ Outline Firewalls 1 Understanding TCP and UDP – Ports – Sockets – User Datagram Protocols – Transmission Control Protocol – Compare TCP and UDP Application Layer Protocols – Hypertext Transfer Protocol – Simple Mail Transfer Control Protocol – File Transfer Protocol – Domain Name Server – Dynamic Host Configuration Protocol Network Security – Needs – History – Four Pillars – Security Terms – Security Components – Types of Attacks – Known Security Attacks – Firewalls Text / Reference: T/R Book Title / Author Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. Elements of T1 Computer Networking: An Integrated Approach. CareerMonk Publications, 2014. Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition, R1 Sybex (Wiley) Publication, 2016 Online Resources: Sl.No. Website Link https://www.tutorialspoint.com/data_communication_computer_network/index.h 1 tm 2 https://www.javatpoint.com/computer-network-tutorial 3 https://www.geeksforgeeks.org/computer-network-tutorials/ □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
protocols and Network Security.

   M4.01       Illustrate Ports and Sockets                    1
Understanding


               Outline Transmission Control Protocol
   M4.02                                                       2
Understanding
               and User Datagram Protocols

   M4.03       Summarize Hypertext Transfer Protocol           2
Understanding
               Outline Simple Mail Transfer Control
   M4.04                                                       1
Understanding
               Protocol
   M4.05       Illustrate File Transfer Protocol               1
Understanding


   M4.06       Summarize Domain Name Server                    1
Understanding
               Outline Dynamic Host Configuration
   M4.07                                                       1
Understanding
               Protocol
               Summarize Network Security Needs,
   M4.08                                                       2
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

## Concept and formula bank

### Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

### Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

## Diagram and source-transcript library

### Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

## Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

## Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

## Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

### Module 1

**Define or explain elements of a computer network and 2 Understanding Network Topologies. (3 marks)**

**Answer:** Begin with the standard definition or identity of *elements of a computer network and 2 Understanding Network Topologies*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Illustrate Transmission Mediums.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Illustrate Transmission Mediums.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Outline Types of Networks. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Outline Types of Networks*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Summarize Switched Networks 2 Understanding Illustrate the duties of each layer in OSI. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Summarize Switched Networks 2 Understanding Illustrate the duties of each layer in OSI*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

## Module 2

**Define or explain 1 Understanding Techniques. (3 marks)**

**Answer:** Begin with the standard definition or identity of *1 Understanding Techniques*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Summarize Ethernet and its types. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Summarize Ethernet and its types*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Outline ARP and RARP. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Outline ARP and RARP*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Summarize IP Address and IP 1 Understanding Addressing Outline Classful IP Addressing and IP. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Summarize IP Address and IP 1 Understanding Addressing Outline Classful IP Addressing and IP*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

## Module 3

**Define or explain 2 Understanding Algorithms Summarize Classification of Routing. (3 marks)**

**Answer:** Begin with the standard definition or identity of *2 Understanding Algorithms Summarize Classification of Routing*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

**Define or explain 2 Understanding Algorithms. (3 marks)**

**Answer:** Begin with the standard definition or identity of *2 Understanding Algorithms*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

**Define or explain Outline Routing Algorithm Metrics. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Outline Routing Algorithm Metrics*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Summarize Internet Architecture 1 Understanding Outline Routing Protocols and its. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Summarize Internet Architecture 1 Understanding Outline Routing Protocols and its*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

## Module 4

**Define or explain Illustrate Ports and Sockets 1 Understanding Outline Transmission Control Protocol. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Illustrate Ports and Sockets 1 Understanding Outline Transmission Control Protocol*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 2 Understanding and User Datagram Protocols. (3 marks)**

**Answer:** Begin with the standard definition or identity of *2 Understanding and User Datagram Protocols*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Summarize Hypertext Transfer Protocol 2 Understanding Outline Simple Mail Transfer Control. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Summarize Hypertext Transfer Protocol 2 Understanding Outline Simple Mail Transfer Control*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 1 Understanding Protocol. (3 marks)**

**Answer:** Begin with the standard definition or identity of *1 Understanding Protocol*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

## Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

### Module 1

1-mark definition: State the meaning of **elements of a computer network and 2 Understanding Network Topologies**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 2

1-mark definition: State the meaning of **1 Understanding Techniques**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 3

1-mark definition: State the meaning of **2 Understanding Algorithms Summarize Classification of Routing**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 4

1-mark definition: State the meaning of **Illustrate Ports and Sockets 1 Understanding Outline Transmission Control Protocol**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

# Exam checklist

### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

# Glossary

**Study glossary**

| Term | Meaning in this lesson |
|---|---|
| Course Outcome | A capability the student should demonstrate after completing the course. |
| Module | An official syllabus unit with defined topics and hours. |
| CIE | Continuous Internal Evaluation. |
| ESE | End Semester Examination. |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

SOURCE-SUPPORTED REFERENCES

# References and web resources

## References listed in the official PDF

References are included in the official source PDF.

### Web resources listed in the official PDF

- Sl.No. Website Link
- https://www.tutorialspoint.com/data_communication_computer_network/index.h
- https://www.javatpoint.com/computer-network-tutorial
- https://www.geeksforgeeks.org/computer-network-tutorials/

IN-PAGE SEARCH

## Search this lesson